



Reducing Simulation Runtime Using Amazon EC2 Hpc7g Instances with Fairbrics

Learn how Fairbrics is accelerating the design of sustainable fabrics out of thin air (carbon dioxide) using Amazon EC2 Hpc7g Instances.

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Reduced

3-month HPC

Cut costs

Optimized



End-to-end

simulation workflow in the cloud

Overview

[Fairbrics](#) invented a new, innovative, and environmentally sustainable method to produce synthetic fabrics, but in order to speed up and optimize the process, Fairbrics required computing power. By using [high performance computing \(HPC\) clusters](#) on Amazon Web Services (AWS), Fairbrics accelerated the simulation time, cut simulation costs, and optimized its compute footprint.



Opportunity | Using Amazon EC2 Hpc7g Instances to Optimize Electrolyzer Simulations for Fairbrics

Founded in 2019, Fairbrics is a chemical manufacturing company that makes fabric out of thin air. It uses waste carbon dioxide captured from industrial sources as a building block to create synthetic polyester textiles, which it then sells to suppliers in the fashion industry. The first step of Fairbrics's innovative multistage process uses an electrolyzer to chemically convert carbon dioxide into carbon monoxide—a necessary component of the manufacturing process.

To be certain the electrolyzer operates efficiently, Fairbrics first runs numerical simulations to understand and improve the flow of liquid and gas inside the electrolyzer. The simulations also help the team design and test new technology to enhance its electrolyzers. These simulations require significant computational resources. A single workstation was enough to model simple test simulations during the early days of the company, but as the simulations became more complex, Fairbrics needed more computing power to maintain efficiency.

In 2023, Fairbrics joined [AWS Activate](#), a program that provides startups with resources to build, launch, and scale on AWS. In addition to providing credits to reduce costs, the program gave Fairbrics support as it started its journey using HPC clusters.

AWS also introduced the Fairbrics team to [UCit](#), founding member of the [Do It Now](#) alliance and an [AWS Partner](#) specializing in HPC. “When we started using AWS, we needed help setting up and interfacing with the clusters to make the simulation process simpler,” says Gregory Pujals, fluid mechanics engineer at Fairbrics. “I started by sharing my issues, and the team offered advice about what to do.” Working alongside UCit, Fairbrics deployed [Amazon Elastic Compute Cloud](#) (Amazon EC2) [Hpc7g Instances](#), HPC-optimized instances that developers use to run compute-intensive workloads such as computational fluid dynamics, weather forecasting, and molecular dynamics. This solution offered Fairbrics greater energy efficiency and price performance, helping it to maintain its commitment to sustainability.

Solution | Reducing Simulation Runtime from Months to Days

From initial engagement to completion, deploying HPC clusters for Fairbrics took 3 months. UCit facilitated the deployment using UCit [CCME](#) (Cloud Cluster Made Easy), which made configuring the HPC clusters simple. After setting up the account, UCit trained the Fairbrics team to launch and use it. “Finding the right solution for Fairbrics was an iterative, conversational process,” says Philippe Bricard, CEO and founder of UCit. “It had to be something that would continue to evolve depending on Fairbrics’s priorities and capabilities going forward.”

Architecture Diagram



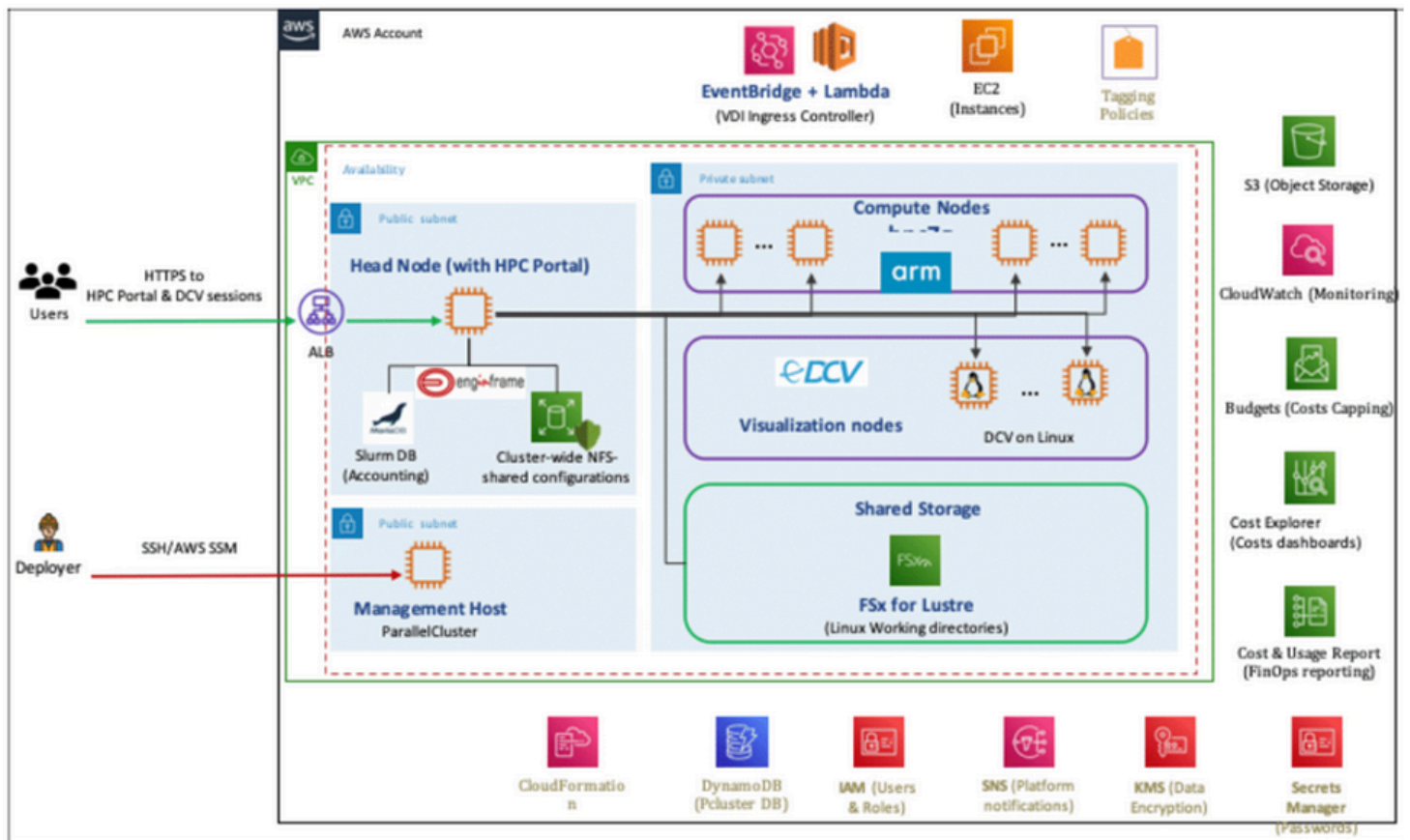


Figure 1. Fairbrics architecture

Amazon EC2 [Hpc7g](#) Instances are powered by [AWS Graviton 3E processors](#), which are designed to deliver the best price performance for cloud workloads running in Amazon EC2. These processors greatly accelerated Fairbrics's simulation runtimes, with one simulation on an HPC cluster being about twice as fast as the time it took previously on a single workstation. Moreover, while a single simulation took 10 hours before using HPC clusters, Fairbrics can now run five simulations in parallel in a few hours. That means the team can get results in days instead of months. "We are developing new kinds of devices, and those have to be tested," says Pujals. "If I can run an accurate simulation and get results in a couple of days, that's valuable."

Using this solution, Fairbrics has achieved better price performance through speed and efficiency; the team can do more with the same budget it always had. At the same time, AWS Graviton 3E processors align with Fairbrics's sustainability goals, which include minimizing climate risk and raising resilience. "It's one of the least energy-hungry technologies," says Bricard. "We knew AWS Graviton 3E processors were the right direction."

UCit also suggested [NICE DCV](#), a high-performance remote display protocol that provides customers with a secure way to deliver remote desktops and application streaming from any cloud or data center to any device, over varying network conditions. This solution makes it possible for Fairbrics to analyze and monitor simulations remotely, which has been useful for the team. It can check that simulations are running correctly anywhere at any time. Another benefit of this solution is security. "We embedded many authentication and security techniques into the solution," says Bricard. "This is absolutely key for Fairbrics because its innovative method of making polyester is a competitive advantage that needs to be kept secure."

Outcome | Creating Innovative Products with AWS HPC Clusters

With a significant boost in efficiency and speed, Fairbrics now has the capability to adapt to its requirements and continue developing sustainable fabric. By using AWS HPC services, the team can create innovative products much faster than before.



“Using AWS empowers us to do more, get more insights, and do our jobs better,” says Pujals. “All the work we’ve done on AWS has helped us to go further than we could before.”

About Fairbrics

Founded in 2019, Fairbrics is a chemical manufacturing company that uses carbon dioxide waste to create synthetic fibers for the textile industry. By making fabric out of thin air, Fairbrics fosters sustainability and reduces emissions.

AWS Services Used

Amazon Elastic Compute Cloud (Amazon EC2) Hpc7g Instances

HPC-optimized instances powered by AWS Graviton3E processors enabling the best price performance for HPC workloads on AWS.

[Learn more »](#)

AWS Graviton 3E Processors

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

NICE DCV

NICE DCV is a high-performance remote display protocol that provides customers with a secure way to deliver remote desktops and application streaming from any cloud or data center to any device, over varying network conditions.

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AWS Activate



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